



Technical note

Calibration of optical particle counters with an aerodynamic aerosol classifier



Nafiseh Sang-Nourpour, Jason S. Olfert*

Department of Mechanical Engineering, University of Alberta, Edmonton, AB, Canada

ARTICLE INFO

Keywords:

Optical particle counter
Calibration
Aerodynamic aerosol classifier
And condensation particle counter

ABSTRACT

A new technique to calibrate the sizing and counting characteristics of single-particle optical particle counters (OPCs) is presented. The method is not limited to a specific material for calibration, rather any material that can be aerosolized may be used for calibration. Examples of aerosols with real and complex refractive indices and spherical and non-spherical shapes are employed to demonstrate this calibration approach. The material choices include bis(2-ethyl-hexyl) sebacate, Fomblin, and soot. To generate monodisperse aerosols, an aerodynamic aerosol classifier is employed along with a constant output atomizer and a miniature inverted soot generator. The optical particle counter is then coupled to a condensation particle counter that facilitates concentration comparisons between the two devices. This approach is useful for OPC calibration for particle sizes less than approximately 5 μm .

1. Introduction

Airborne particulate matter has notable impacts on human health, the environment, and climate change (Anderson, 2009; Landen, Wassell, McWilliams, & Patel, 2011; Li, Guo, & Williams, 2016; Shah et al., 2013; Tittarelli et al., 2008). Many aerosol instruments including, but not limited to, the scanning mobility particle sizer (Wang & Flagan, 1990), aerodynamic particle sizer (Baron, 1986) and optical particle counters/sizers (OPCs) (Sorensen, Gebhart, O'Hern, & Rader, 2011), are commonly used to determine the size and concentration of airborne particles in various applications including in atmospheric measurements, occupational assessments, laboratories, clean rooms, etc. In contrast to other devices, OPCs are commonly available in portable, light-weight models and are generally lower cost compared to other particle sizing devices (Sousan, Koehler, Hallett, & Peters, 2016). Recently, low-cost sensors have become available on the market that are sufficiently inexpensive that they can be deployed in vast numbers to measure particulate concentrations with high spatial and temporal resolution (Manikonda, Zíková, Hopke, & Ferro, 2016).

Single particle optical particle counters measure the concentration and size of individual particles by employing light scattering (Sorensen et al., 2011; Welker, 2012). They have applications in environmental and air pollution studies (Bezantakos, Schmidt-Ott, & Biskos, 2018; Görner, Simon, Bémer, & Lidén, 2012; Jurányi, Bartscher, Loepfe, Nenkov, & Weingartner, 2015; Kobza, Geremek, & Dul, 2018), pharmaceutical analysis and basic aerosol research (Iida & Sakurai, 2018; Mirhoseini, Nikaeen, Khanahmd, Hatamzadeh, & Hassanzadeh, 2015). Commercially available OPCs are able to size particles in a vast size range from 50 nm to hundreds of micrometers, although sizing over this range is generally not done with one device (Welker, 2012).

Different commercial implementations of single particle OPCs collect the scattered light at different angles (e.g. 90° or forward scattered light). The scattered light intensity depends on particle size, refractive index, and scattering angle (Sorensen et al., 2011). In

* Corresponding author.

E-mail address: jolfert@ualberta.ca (J.S. Olfert).

general, there are two types of single particle OPCs: *ex-situ* or *in-situ*. *Ex-situ* OPCs bring the aerosol into an optical chamber and depending on the application, different light illumination and collection angles and flow configurations have been employed (Chen, Cheng, & Yeh, 1984; Sorensen et al., 2011). *In-situ* OPCs (or open-path OPCs) have external optics, the particles are not sampled into the instrument, rather the instrument measures directly from the ambient air (Baumgardner, Jonsson, Dawson, O'Connor, & Newton, 2001; Bu, Yuan, Huang, & Chang, 2013; Cantin, 2018; Dean, 2014; Rosenberg et al., 2012).

The response of the OPC to various particle sizes and materials can be theoretically computed, however, the theory does not compensate for differences in the manufacturing of each instrument. Therefore, instruments are usually calibrated by the manufacturer or user before being used (Sorensen et al., 2011). Calibration of OPCs is also used in some circumstances to find the response of the instrument to particles of different refractive index and morphology (Marple & Rubow, 1976; Szymanski & Liu, 1986).

Optical particle counters are often calibrated in terms of the particle size and number concentration they measure (e.g. Sorensen et al. (2011)). Other techniques have been employed to calibrate OPCs in terms of mass concentration; which is useful for low-cost optical sensors designed for measuring particle mass concentration (Collingwood et al., 2019; Liu, Zhang, Jiang, & Chen, 2017; Manikonda et al., 2016).

Various calibration approaches have been used for *ex-situ* OPC calibration in terms of particle size and number concentration. Marple and Rubow (1976) calibrated an OPC with respect to aerodynamic particle size using inertial impactors. In this system, two measurements of a calibration aerosol are made with the OPC — one with an impactor on the inlet of the OPC and one without. The optical cut-off diameter measured by the OPC is compared to the cut-off aerodynamic diameter of the impactor to obtain a single calibration point. Different calibration points are attained by changing the cut-off diameter of the impactor (Marple & Rubow, 1976). Similar techniques have been used with other pre-collectors on the inlet of the OPC (Fissan, Helsper, & Kasper, 1984) including cyclones (Büttner, 1983) and partially-penetrating filters (Liebhaber & Willeke, 1993).

OPCs have also been calibrated by comparing the particle size distribution (or cumulative distribution) measured by the OPC to the size distribution measured in some other way. This has included comparisons to electron microscopy measurements (Liu, Marple, Whitby, & Barsic, 1974b) and aerodynamic particle sizers either with a pre-collection device (Binnig, Meyer, & Kasper, 2007) or without (Chien, Theodore, Wu, Hsu, & Birky, 2016; Friehmelt & Heidenreich, 1999). Comparisons have also been made using a scanning mobility particle sizer for particles less than $1\ \mu\text{m}$ in diameter (Sousan et al., 2016).

OPC calibration is also done using monodisperse test aerosols with known size and refractive index, such as polystyrene latex spheres (PSL) (Sorensen et al., 2011). In this case the PSL is placed in solution, atomized, dried, and measured by the OPC. Monodisperse particles of other materials may be made with monodisperse droplet generators such as vibrating orifice aerosol generators, which produce monodisperse aerosols by changing the vibration frequency and orifice liquid feed (Berglund & Liu, 1973; Chen et al., 1984; Liu, Berglund, & Agarwal, 1974a), condensation-type aerosol generators (Belosi, Santachiara, & Prodi, 2013), or inkjet aerosol generators (Iida & Sakurai, 2018). The differential mobility analyzer (DMA, Knutson and Whitby (1975)), which classifies particles based on their electric mobility, has also been used to calibrate OPCs (Heim, Mullins, Umhauer, & Kasper, 2008). However, the DMA is generally used for particle sizes smaller than $\sim 1\ \mu\text{m}$, because for larger sizes, the increased probability of classifying multiply-charged particles may be problematic if the OPC does not have sufficient resolution to resolve the different particle sizes.

There are challenges regarding the existing calibration approaches. The approaches based on pre-collectors (e.g. impactors, cyclones) allow calibration with any material but may take a considerable amount of time to change the pre-collector to get different cutoffs. The microscopy approach is very time consuming and is rarely used for calibration. PSL and the vibrating orifice aerosol generator approach to generate monodisperse aerosols is limited to certain particle materials.

Developing a technique to calibrate *ex-situ* OPCs that is not time consuming and can be used with any aerosol (regardless of material or shape) is advantageous. Here, a new technique is demonstrated to calibrate OPCs in terms of particle size and concentration, which is not limited to a specific particle material or aerosol generation method. Here, an aerodynamic aerosol classifier (AAC) is used to classify a polydisperse aerosol into a monodisperse aerosol for OPC size calibration. The AAC classifies particles based on their aerodynamic diameter over a relatively wide range (approximately $25\ \text{nm}$ to $5\ \mu\text{m}$), independent of particle charge state, with high transmission efficiency (Tavakoli, Symonds, & Olfert, 2014; Tavakoli & Olfert, 2013). In this technique, a polydisperse aerosol is classified by the AAC and then the classified aerosol is measured by the OPC and a condensation particle counter (CPC). The AAC is used to calibrate the OPC in terms of size and the CPC is used to calibrate the OPC in terms of number concentration. Here we demonstrate this calibration technique with spherical particles of known refractive index (bis(2-ethylhexyl) sebacate and Fomblin) and soot, which has an irregular shape and complex refractive index. The limitations of using the CPC for calibrating the OPC for large particle sizes is also investigated.

2. Experimental setup

The schematic of the setup for calibrating an OPC in terms of size and concentration is presented in Fig. 1. The setup includes the calibration system (AAC, CPC, and OPC) in Fig. 1a and aerosol generators (atomizer or soot generator) in Fig. 1b and c.

Materials with real and complex refractive indices were tested, including bis(2-ethylhexyl) sebacate (DOS), Fomblin, PSL, and soot. The DOS, Fomblin, and PSL represent spherical particles with a wide range of densities and refractive indices as shown in Table 1. DOS, Fomblin and PSL particles were generated with a constant output atomizer (TSI Inc., Model 3076) as shown in Fig. 1b. The DOS (Aldrich Chemistry, purity >97%, CAS# 122-62-3) and Fomblin (Solvay, Fomblin Y L-VAC 25/6, 33334SA, CAS# 69991-67-9) form spherical aerosols with near-zero absorption. PSL monodisperse spheres with diameters of $400\pm 9\ \text{nm}$ and $1000\pm 100\ \text{nm}$ (Thermo-Scientific, Nanosphere 3400A; Magsphere Inc., Monosized PS Microspheres PS001UM), were also used as a reference check

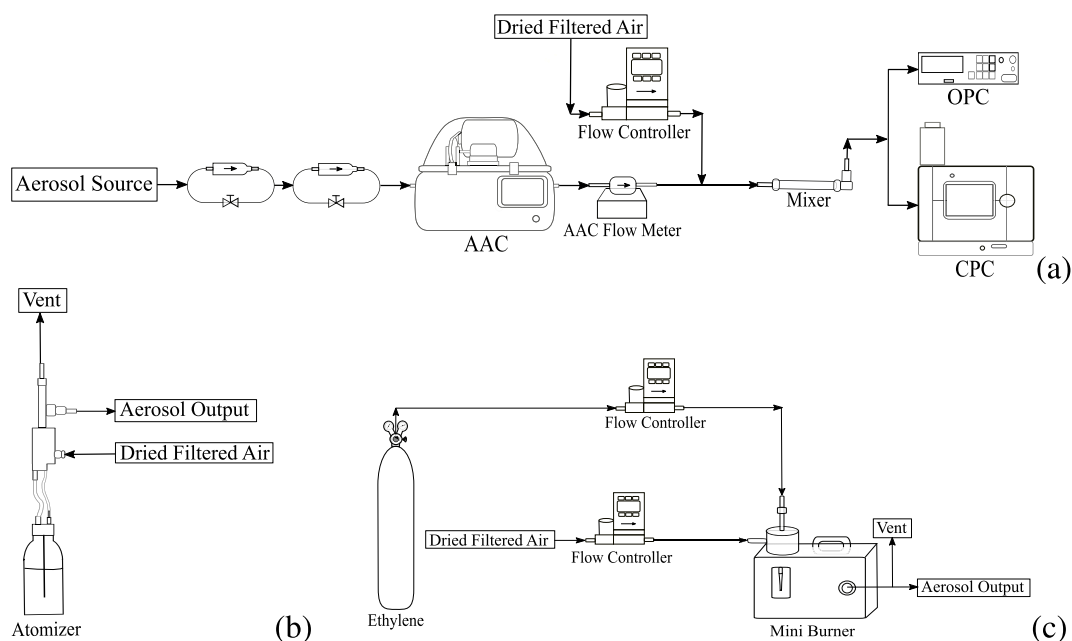


Fig. 1. (a) Setup schematic for the calibration of an OPC with an AAC and CPC. The aerosol sources are (b) a constant output atomizer for DOS, Fomblin, and PSL and (c) miniature inverted soot generator for soot.

Table 1

The refractive index and material density of DOS, Fomblin, PSL, and soot.

Material	Refractive index (at 20°C, 589 nm)	Material density (kg/m ³)
DOS (Lide, 1998-1999)	1.45	912
Fomblin (Solvay; Sigma-Aldrich)	1.3	1900
PSL (Davis & Ravindran, 1982)	1.59	1050
Soot (Bond & Bergstrom, 2006; Park, Kittelson, Zachariah, & McMurry, 2004)	1.95–0.79i	1770

for the OPC.

Soot was generated by employing a miniature inverted soot generator (Argonaut Scientific) as shown in Fig. 1c. The operating principle of the soot generator is described in detail in by Kazemimanesh et al. (2019) and Moallemi et al. (2019). Ethylene and dry, filtered, compressed air were provided to the burner with mass flow controllers (Alicat, MC-200SCCM-D/5M and MC-20SLPM-D/5M, respectively). The ethylene flow rate was set to 0.098 standard L/min and the air flow rate was 10 standard L/min (reference temperature and pressure of 25°C and 101.325 kPa, respectively). The output of the soot generator was diluted with an ejector diluter (Dekati, DI-1000) by a ratio of approximately 8:1 to reduce soot concentration.

In the calibration setup (Fig. 1a), aerosols generated by the constant output atomizer or miniature inverted soot generator passed through two HEPA filter-needle valve bridges to facilitate the control of the aerosol concentration going into the AAC. The aerosol was then classified by aerodynamic diameter by the AAC. The AAC (Cambustion Ltd) classifies aerosols based on their aerodynamic diameter (or particle relaxation time). The AAC output is a monodisperse aerosol with a size distribution width that is proportional to the AAC resolution parameter. The AAC resolution is defined as the inverse of the full-width half-maximum of the AAC transfer function. The AAC resolution was set to 10 in terms of aerodynamic diameter for all measurements (the AAC software controls the sheath flow to obtain the desired resolution for each particle size (Tavakoli & Olfert, 2013)). The sample flow rate through the AAC was measured by the AAC flow meter (Cambustion Ltd.) and was controlled by adding dry, filtered, compressed air with a mass flow controller (Cole-Parmer, 32907-71) after the AAC flow meter. The flow controller was set so that the flow rate through the AAC was ~0.45 L/min.

The diluted flow was then drawn through a static mixer (Ko-flo Corporation, 1/4-40-2-6-2) to ensure the classified aerosol and air were completely mixed before the flow was split to the CPC and OPC. The CPC (TSI Inc., Model 3776) measured the aerosol concentration and the inlet flow rate of the CPC was set to 1.5 L/min.

The OPC in the experiments was the Grimm 11-C portable aerosol spectrometer that detects aerosols in the size range of 250 nm–32 µm with a concentration up to approximately $2 \times 10^3 \text{ cm}^{-3}$. The Grimm 11-C detects scattered light from particles over the angular range of 60°–120°. The OPC averaging time was set to 6 s and the inlet flow rate was 1.21 L/min. The line lengths and radii of curvature of the conductive tubing to the CPC and OPC were identical to minimize the difference in the particle losses due to inertial impaction and diffusion in the two sampling lines.

For each aerosol source, the AAC was stepped through a range of sizes. The lower size limit was set to be near the lower detection limit of the OPC (~250 nm). The upper limit was determined by the aerosol source. The AAC can classify particle up to ~5 μm ; however, the particle sources did not generate high concentrations of large particles. The largest sizes that could be produced for each source were $d_a=2500$ nm, 2500 nm and 700 nm; for DOS, Fomblin, and soot; respectively. At each particle size set by the AAC, the filter bridges were adjusted to provide a range of aerosol concentrations.

3. Data analysis

3.1. Particle diameters

The AAC classifies particles based on their aerodynamic equivalent diameter; however, results will also be presented in terms of volume equivalent diameter. For spherical particles (DOS, Fomblin or PSL), the volume equivalent diameter is (DeCarlo, Slowik, Worsnop, Davidovits, & Jimenez, 2004),

$$d_{ve} = d_a \sqrt{\frac{\rho_0 C_c(d_a)}{\rho_p C_c(d_{ve})}}, \quad (1)$$

where, C_c is the slip correction factor as a function of the aerodynamic or volume equivalent diameter, ρ_p is the material density and ρ_0 is a reference density ($\rho_0 = 1000 \text{ kg/m}^3$).

For non-spherical particles the volume equivalent diameter can be related to aerodynamic diameter if the effective density (ρ_{eff}) of the particle is known, where the effective density is the mass of the particle divided by its mobility equivalent volume. Thus, the volume equivalent diameter is,

$$d_{ve} = d_a \sqrt{\frac{\rho_0}{\rho_p^{\frac{2}{3}} \rho_{eff}^{\frac{1}{3}}} \frac{C_c(d_a)}{C_c(d_m)}}, \quad (2)$$

where, d_m is the mobility diameter. For soot particles $\rho_p = 1770 \text{ kg/m}^3$ (Park et al., 2004) and the effective density is estimated to be,

$$\rho_{eff} = (510 \text{ kg/m}^3) \left(\frac{d_m}{100 \text{ nm}} \right)^{-0.52}, \quad (3)$$

based on the measurements of Olfert and Rogak (2019) for a wide range of soot sources.

3.2. Data processing

The particles exiting the AAC will have a narrow range of aerodynamic diameters which were measured by the OPC. Examples of DOS size distributions measured by the OPC for three aerodynamic diameters are shown in Fig. 2. The optical diameter was compared to the aerodynamic diameter by comparing the geometric mean of the optical diameter distribution to the set point of the AAC (i.e. the aerodynamic diameter that corresponds to the maximum of the AAC transfer function). As shown in the figure, the peaks of the OPC optical diameter distribution are consistently lower than the aerodynamic diameter of the particles indicating that the OPC is under-sizing the DOS particles.

For only large sizes of DOS and Fomblin particles (> 1500 nm), the OPC detected high particle concentrations in the smallest size bins (optical diameters less than ~300 nm). An example of this artifact is shown in Fig. 2 for the size distributions of DOS aerosols for AAC sizes set to 1800 nm and 2500 nm. A scanning mobility particle sizer (SMPS) was used in parallel with the OPC to determine if

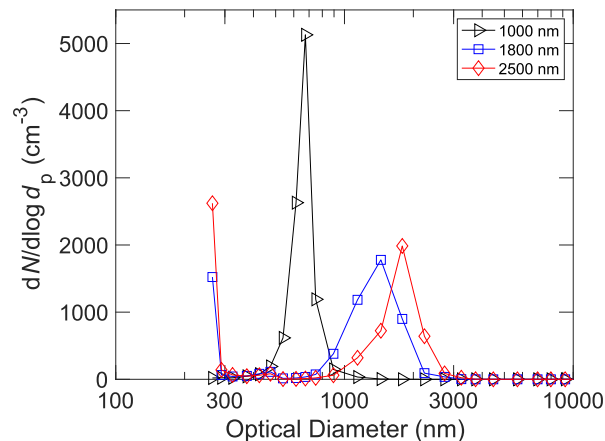


Fig. 2. The average OPC size distributions for 1000 nm, 1800 nm, and 2500 nm DOS aerosols.

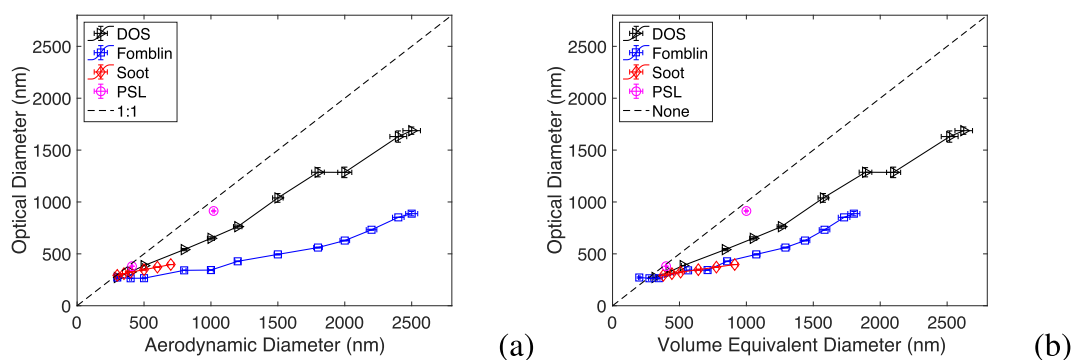


Fig. 3. OPC optical diameter versus (a) aerodynamic diameter and (b) volume equivalent diameter for DOS, Fomblin, soot and PSL aerosols.

this was an artifact of the OPC measurement or perhaps due to mis-classification of the AAC. The SMPS determined there were no particles with diameters below 700 nm, thus, we concluded that the small particles detected by the OPC were an artifact of the OPC measurement. Belosi et al. (2013) found a similar artifact for the Grimm 1.108 OPC measuring 1 μm and 1.4 μm diameter PSL and carnauba wax particles. Thus, when this artifact was observed, the particles measured in the smallest size bins were removed from subsequent data analysis.

4. Results and discussion

The optical diameters of DOS, Fomblin and soot detected by the OPC versus AAC-classified aerodynamic and volume equivalent diameters are presented in Fig. 3. The error bars on the y-axis represent precision uncertainty of the OPC size with 95% confidence. The x-axis error bars for the aerodynamic diameter represent the AAC accuracy, which is estimated to be $\pm 5\%$ (Johnson, Irwin, Symonds, Olfert, & Boies, 2018), while the error bars for the volume equivalent diameter are calculated using the method described in Appendix A.

For reference, the OPC was also tested with monodisperse PSL (without AAC classification) at 400 nm and 1000 nm (volume equivalent diameter), since the OPC was factory-calibrated with PSL. As shown in Fig. 3b, the OPC optical diameter is close to the volume equivalent diameter of the PSL demonstrating that the OPC sizes PSL particles as expected.

As shown in Fig. 3, by comparing the measurements between DOS, Fomblin, soot and PSL, it is apparent that OPC sizing is dependent on the particle refractive index (see Table 1). This can be seen by comparing the spherical particles, where PSL has the highest refractive index and largest optical diameter, while Fomblin has the lowest refractive index and smallest optical diameter for a given volume equivalent diameter, as expected from Mie theory (Hinds, 1999).

Also, note that in the case of soot and Fomblin there is a large difference between their respective volume equivalent and aerodynamic diameters due to their density or morphology. This results in very different calibration curves depending on which equivalent diameter is used. For example, soot and Fomblin have similar responses in the OPC in term of volume-equivalent diameter (Fig. 3b), but quite different responses in terms of aerodynamic diameter (Fig. 3a).

The OPC concentration was compared to the concentration detected by CPC to validate the OPC accuracy in detecting the aerosol number concentration. Fig. 4 shows the concentration measurements of the OPC and CPC for aerodynamic diameters of 300 nm, 1000 nm, and 2000 nm of DOS particles as an example. The figure shows that there is a linear relationship between the OPC and the CPC but it is a function of the particle size. At small sizes of aerosols, the OPC under-counts the aerosol concentration while at

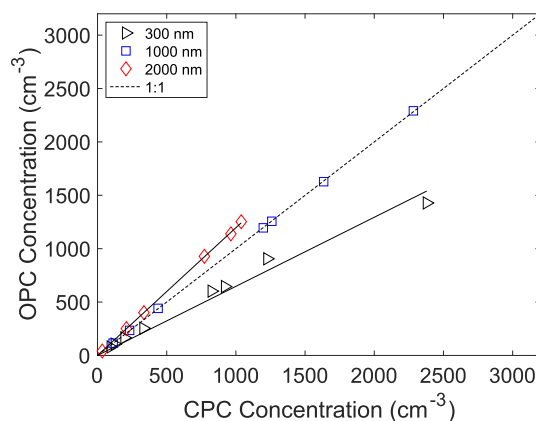


Fig. 4. The concentrations of OPC versus CPC for selected sizes of DOS for 300 nm, 1000 nm, and 2000 nm DOS aerosols.

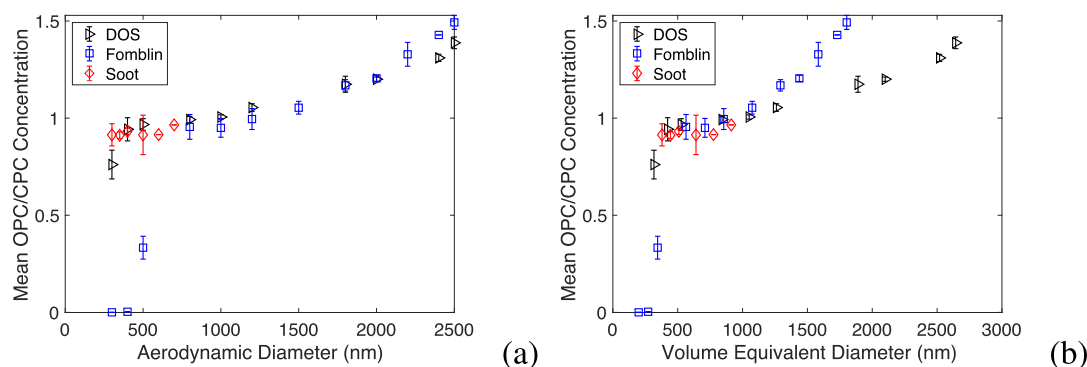


Fig. 5. Relative concentration of OPC to CPC by changing (a) aerodynamic diameter and (b) volume equivalent diameter for DOS, Fomblin and soot aerosols. The vertical error bars represent the precision uncertainty with 95% confidence.

aerodynamic diameters larger than ~ 1500 nm, the CPC under-counts relative to the OPC.

The relationship between the counting efficiencies of the instruments is more clearly seen by plotting the ratio of the OPC to CPC concentrations as a function of the particle size as shown in Fig. 5 in terms of aerodynamic diameter (Fig. 5a) and volume equivalent diameter (Fig. 5b) for all particle types. In Fig. 5b, the OPC under-counts at lower sizes because the OPC is limited by the software to only report optical sizes larger than 250 nm, while the CPC has nearly 100% detection efficiency for particles in the range 3 nm to approximately 1 or 2 μm . Thus, the OPC detection efficiency has a relatively sharp cut-off at lower particle sizes. Moreover, note that the AAC transfer function has a finite width, thus some particles exiting the AAC may be above or below the OPC size limit and may or may not be detected by the OPC resulting in a fraction of the particles being detected at given AAC set point.

At sizes greater than ~ 1500 nm in aerodynamic diameter, the OPC over-counted relative to CPC. This is likely not due to errors in the OPC, but rather due to poor counting efficiency of the CPC. The counting efficiency at large particle sizes is not known for the Model 3776 CPC. However, the Model 3776 CPC has a relatively complex flow path from the inlet to the growth chamber where it might be expected that large particles could be lost. These losses are expected to be inertial, and therefore, a function of the aerodynamic size of the particles. Indeed, by inspection of Fig. 5a one can see that the concentration ratios of DOS and Fomblin are very similar for large particle sizes in terms of aerodynamic diameter, suggesting the difference in the counting efficiencies of the instruments is due to inertial particle losses within the CPC. Therefore, if this calibration technique is to be used to calibrate the counting efficiency of the OPC at large particle sizes, then a CPC is required which has high counting efficiency in the micron size range. The detection efficiency of a CPC up to diameters of $\sim 5 \mu\text{m}$ can be determined with the technique demonstrated by Järvinen, Keskinen, and Yli-Ojanperä (2018).

5. Conclusion

The material-dependence of OPC sizing is well known and is a major motivation for OPC calibration. A method to calibrate an OPC in terms of size and concentration for different aerosol materials was presented. Employing this technique, OPCs can be calibrated with any material that can be atomized. This technique was demonstrated for particle sizes up to $2.5 \mu\text{m}$ due to limitations in the particle generation methods used here; however, sizes up to $\sim 5 \mu\text{m}$ are possible with the AAC. If a CPC is used to calibrate the OPC in terms of number concentration, then care must be taken to ensure the counting efficiency of the CPC is near 100% over the particle size range of interest.

Although the purpose of the technique demonstrated here was to calibrate the size and number concentration of an OPC, the technique could be extended to calibrate an OPC in terms of mass concentration (as a function of particle size) if the mass of individual AAC-classified particles is known (*i.e.*, the mass concentration will be the mass of individual particles multiplied by the number concentration measured by the CPC). For spherical particles the mass of the particle is known if the particle density is known. For all other particles, the particle mass could be measured with a particle mass classifier such as an aerosol particle mass analyzer (APM; Ehara, Hagwood, and Coakley (1996)) or centrifugal particle mass analyzer (CPMA; Olfert and Collings (2005)).

Acknowledgement

Funding was provided by the Natural Sciences and Engineering Research Council of Canada. We thank Dr. Sarah Styler of the University of Alberta for lending the OPC used in this work.

Appendix A. Uncertainty estimation for measured and derived parameters

This appendix shows how the uncertainties were calculated for the calculation of volume equivalent diameter from the experimental data. As shown in Eq. (1), the volume equivalent diameter for spherical particles (DOS, Fomblin or PSL) is,

$$d_{ve} = d_a \sqrt{\frac{\rho_0 C_c(d_a)}{\rho_p C_c(d_{ve})}}. \quad (A.1)$$

Using the propagation of error, the uncertainty in the volume equivalent diameter is

$$\left(\frac{\varepsilon_{d_{ve}}}{d_{ve}}\right)^2 = \left(\frac{\varepsilon_{d_a}}{d_a}\right)^2 + \frac{1}{4}\left(\frac{\varepsilon_{\rho_p}}{\rho_p}\right)^2 + \frac{1}{4}\left(\frac{\varepsilon_{C_c(d_a)}}{C_c(d_a)}\right)^2 + \frac{1}{4}\left(\frac{\varepsilon_{C_c(d_{ve})}}{C_c(d_{ve})}\right)^2, \quad (A.2)$$

where $\varepsilon_{d_a}/d_a = 5\%$ (Johnson et al., 2018). It is assumed that the uncertainty in slip correction factors is the same for all particle sizes and is $\pm 2.1\%$ (Allen & Raabe, 1985). It is assumed the uncertainty of the material density of each material is $\pm 10 \text{ kg/m}^3$. Therefore, the uncertainty in the volume equivalent diameter for DOS is

$$\frac{\varepsilon_{d_{ve}}}{d_{ve}} = \sqrt{(0.05)^2 + \frac{1}{4}\left(\frac{10 \text{ kg/m}^3}{912 \text{ kg/m}^3}\right)^2 + \frac{1}{4}(0.021)^2 + \frac{1}{4}(0.021)^2} = 0.0524, \quad (A.3)$$

which corresponds to $\pm 5.24\%$ uncertainty. Likewise, for Fomblin and PSL the uncertainty will be $\pm 5.22\%$ and $\pm 5.23\%$; respectively.

As shown in Eq. (2), the volume equivalent diameter of soot can be calculated as,

$$d_{ve} = d_a \sqrt{\frac{\rho_0}{\rho_p^{\frac{2}{3}} \rho_{eff}^{\frac{1}{3}}} \frac{C_c(d_a)}{C_c(d_m)}}. \quad (A.4)$$

Using the propagation of error, the uncertainty in the volume equivalent diameter is,

$$\begin{aligned} \left(\frac{\varepsilon_{d_{ve}}}{d_{ve}}\right)^2 &= \left(\frac{\varepsilon_{d_a}}{d_a}\right)^2 + \frac{1}{9}\left(\frac{\varepsilon_{\rho_p}}{\rho_p}\right)^2 + \frac{1}{36}\left(\frac{\varepsilon_{\rho_{eff}}}{\rho_{eff}}\right)^2 \\ &+ \frac{1}{4}\left(\frac{\varepsilon_{C_c(d_a)}}{C_c(d_a)}\right)^2 + \frac{1}{4}\left(\frac{\varepsilon_{C_c(d_{ve})}}{C_c(d_{ve})}\right)^2. \end{aligned} \quad (A.5)$$

It is assumed that the uncertainty in the soot material density, ε_{ρ_p} , is $\pm 70 \text{ kg/m}^3$ (Park et al., 2004) and the uncertainty in the effective density is $\pm 20\%$ (Olfert & Rogak, 2019). Therefore, the uncertainty in the soot volume equivalent diameter is

$$\begin{aligned} \frac{\varepsilon_{d_{ve}}}{d_{ve}} &= \sqrt{(0.05)^2 + \frac{1}{9}\left(\frac{70 \text{ kg/m}^3}{1770 \text{ kg/m}^3}\right)^2 + \frac{1}{36}(0.20)^2 + \frac{1}{4}(0.021)^2 + \frac{1}{4}(0.021)^2} \\ &= 0.063, \end{aligned} \quad (A.6)$$

which corresponds to $\pm 6.3\%$ uncertainty.

References

- Allen, M. D., & Raabe, O. G. (1985). Slip correction measurements of spherical solid aerosol particles in an improved millikan apparatus. *Aerosol Science and Technology*, 4(3), 269–286.
- Anderson, H. R. (2009). Air pollution and mortality: A history. *Atmospheric Environment*, 43, 142–152.
- Baron, P. A. (1986). Calibration and use of the aerodynamic particle sizer (APS 3300). *Aerosol Science and Technology*, 5(1), 55–67.
- Baumgardner, D., Jonsson, H., Dawson, W., O'Connor, D., & Newton, R. (2001). The cloud, aerosol and precipitation spectrometer: A new instrument for cloud investigations. *Atmospheric Research*, 59–60, 251–264.
- Belosi, F., Santachiara, G., & Prodi, F. (2013). Performance evaluation of four commercial optical particle counters. *Atmospheric and Climate Sciences*, 3(1), 41–46.
- Berglund, R. N., & Liu, B. Y. H. (1973). Generation of monodisperse aerosol standards. *Environmental Science and Technology*, 7(2), 147–153.
- Bezantakos, S., Schmidt-Ott, F., & Biskos, G. (2018). Performance evaluation of the cost-effective and lightweight Alphasense optical particle counter for use onboard unmanned aerial vehicles. *Aerosol Science and Technology*, 52(4), 385–392.
- Binnig, J., Meyer, J., & Kasper, G. (2007). Calibration of an optical particle counter to provide PM2.5 mass for well-defined particle materials. *Journal of Aerosol Science*, 38(3), 325–332.
- Bond, T. C., & Bergstrom, R. W. (2006). Light absorption by carbonaceous particles: An investigative review. *Aerosol Science and Technology*, 40(1), 27–67.
- Büttner, H. (1983). Kalibrierung einer streulicht-messeinrichtung zur partikelgrößenanalyse mit impaktoren. *Chemie Ingenieur Technik*, 55, 65–76.
- Bu, L., Yuan, J., Huang, X., & Chang, Y. (2013). A cloud droplet probe based on forward scattering. *Optik*, 124(22), 5878–5881.
- Cantin, D. (2018). System and method for individual particle sizing using light scattering techniques. US Patent App. 15/764,833.
- Chen, B. T., Cheng, Y. S., & Yeh, H. C. (1984). Experimental responses of two optical particle counters. *Journal of Aerosol Science*, 15(4), 457–464.
- Chien, C. H., Theodore, A., Wu, C. Y., Hsu, Y. M., & Birky, B. (2016). Upon correlating diameters measured by optical particle counters and aerodynamic particle sizers. *Journal of Aerosol Science*, 101, 77–85.
- Collingwood, S., Zmoos, J., Pahl, L., Wong, B., Sleeth, D., & Handy, R. (2019). Investigating measurement variation of modified low-cost particle sensors. *Journal of Aerosol Science*, 135, 21–32.
- Davis, E. J., & Ravindran, P. (1982). Generation of monodisperse aerosol standards. *Aerosol Science and Technology*, 1(3), 337–350.
- Dean, A. (2014). *Calibration of a cloud droplet probe: A review* Master of Philosophy Thesis. The School of Earth, Atmospheric and Environmental Sciences, University of Manchester.
- DeCarlo, P. F., Slowik, J. G., Worsnop, D. R., Davidovits, P., & Jimenez, J. L. (2004). Particle morphology and density characterization by combined mobility and aerodynamic diameter measurements. Part 1: theory. *Aerosol Science and Technology*, 38(12), 1185–1205.
- Ehara, K., Hagwood, C., & Coakley, K. J. (1996). Novel method to classify aerosol particles according to their mass-to-charge ratio—aerosol particle mass analyser. *Journal of Aerosol Science*, 27, 217–234.
- Fissan, H., Helsper, C., & Kasper, W. (1984). Calibration of optical particle counters with respect to particle size. *Particle & Particle Systems Characterization*, 1(1–4), 108–111.
- Friehmelt, R., & Heidenreich, S. (1999). Calibration of a white light/90° optical particle counter for “aerodynamic” size measurements—experiments and calculations

- for spherical particles and Quartz dust. *Journal of Aerosol Science*, 30(10), 1271–1279.
- Görner, P., Simon, X., Bémer, D., & Lidén, G. (2012). Workplace aerosol mass concentration measurement using optical particle counters. *Journal of Environmental Monitoring*, 14, 420–428.
- Heim, M., Mullins, B. J., Umhauer, H., & Kasper, G. (2008). Performance evaluation of three optical particle counters with an efficient “multimodal” calibration method. *Journal of Aerosol Science*, 39(12), 1019–1031.
- Hinds, W. C. (1999). *Aerosol technology: Properties, behavior, and measurement of airborne particles* (2nd ed.). United States of America: John Wiley & Sons Inc.
- Iida, K., & Sakurai, H. (2018). Counting efficiency evaluation of optical particle counters in micrometer range by using an inkjet aerosol generator. *Aerosol Science and Technology*, 52(10), 1156–1166.
- Järvinen, A., Keskinen, J., & Yli-Ojanperä, J. (2018). Extending the Faraday cup aerosol electrometer based calibration method up to 5 μm . *Aerosol Science and Technology*, 52(8), 828–840.
- Johnson, T. J., Irwin, M., Symonds, J. P. R., Olfert, J. S., & Boies, A. M. (2018). Measuring aerosol size distributions with the aerodynamic aerosol classifier. *Aerosol Science and Technology*, 52(6), 655–665.
- Jurányi, Z., Bartscher, H., Loepe, M., Nenkov, M., & Weingartner, E. (2015). Dual-wavelength light-scattering technique for selective detection of volcanic ash particles in the presence of water droplets. *Atmospheric Measurement Technology*, 8(12), 5213–5222.
- Kazemimanes, M., Moallemi, A., Thomson, K., Smallwood, G., Lobo, P., & Olfert, J. S. (2019). A novel miniature inverted-flame burner for the generation of soot nanoparticles. *Aerosol Science and Technology*, 53(2), 184–195.
- Knutson, E. O., & Whitby, K. T. (1975). Aerosol classification by electric mobility: Apparatus, theory, and applications. *Journal of Aerosol Science*, 8, 443–451.
- Kobza, J., Geremek, M., & Dul, L. (2018). Characteristics of air quality and sources affecting high levels of PM₁₀ and PM_{2.5} in Poland, Upper Silesia urban area. *Environmental Monitoring and Assessment*, 190(9), 515.
- Landen, D. D., Wassell, J. T., McWilliams, L., & Patel, A. (2011). Coal dust exposure and mortality from ischemic heart disease among a cohort of U.S. coal miners. *American Journal of Industrial Medicine*, 54(10), 727–733.
- Lide, D. (Ed.). (1998-1999). *CRC handbook of Chemistry and physics* (79th ed.). Boca Raton, FL: CRC Press Inc.
- Liebhaber, F., & Willeke, K. (1993). Optical particle sizing: Filtering out refractive index effect. *Annals of Occupational Hygiene*, 37(4), 377–394.
- Li, S., Guo, Y., & Williams, G. (2016). Acute impact of hourly ambient air pollution on preterm birth. *Environmental Health Perspectives*, 124(10), 1623–1629.
- Liu, B. Y., Berglund, R. N., & Agarwal, J. K. (1974a). Experimental studies of optical particle counters. *Atmospheric Environment*, 8(7), 717–732.
- Liu, B. Y., Marple, V. A., Whitby, K. T., & Barsic, N. J. (1974b). Size distribution measurement of airborne coal dust by optical particle counters. *American Industrial Hygiene Association Journal*, 35(8), 443–451.
- Liu, D., Zhang, Q., Jiang, J., & Chen, D.-R. (2017). Performance calibration of low-cost and portable particulate matter (PM) sensors. *Journal of Aerosol Science*, 112, 1–10.
- Manikonda, A., Zíková, N., Hopke, P. K., & Ferro, A. R. (2016). Laboratory assessment of low-cost PM monitors. *Journal of Aerosol Science*, 102, 29–40.
- Marple, V. A., & Rubow, K. L. (1976). Aerodynamic particle size calibration of optical particle counters. *Journal of Aerosol Science*, 7(5), 425–428.
- Mirhoseini, S. H., Nikaen, M., Khanahmd, H., Hatamzadeh, M., & Hassanzadeh, A. (2015). Monitoring of airborne bacteria and aerosols in different wards of hospitals – particle counting usefulness in investigation of airborne bacteria. *Annals of Agricultural and Environmental Medicine*, 22(4), 670–673.
- Moallemi, A., Kazemimanes, M., Corbin, J. C., Thomson, K., Smallwood, G., Olfert, J. S., et al. (2019). Characterization of black carbon particles generated by a propane-fueled miniature inverted soot generator. *Journal of Aerosol Science*, 135, 46–57.
- Olfert, J., & Collings, N. (2005). New method for particle mass classification—the Couette centrifugal particle mass analyzer. *Journal of Aerosol Science*, 36(11), 1338–1352.
- Olfert, J., & Rogak, S. (2019). Universal relations between soot effective density and primary particle size for common combustion sources. *Aerosol Science and Technology*, 53(5), 485–492.
- Park, K., Kittelson, D. B., Zachariah, M. R., & McMurtry, P. H. (2004). Measurement of inherent material density of nanoparticle agglomerates. *Journal of Nanoparticle Research*, 6(2), 267–272.
- Rosenberg, P. D., Dean, A. R., Williams, P. I., Dorsey, J. R., Minikin, A., Pickering, M. A., et al. (2012). Particle sizing calibration with refractive index correction for light scattering optical particle counters and impacts upon PCASP and CDP data collected during the Fennec campaign. *Atmospheric Measurement Technology*, 5(5), 1147–1163.
- Shah, A. S., Langrish, J. P., Nair, H., McAllister, D. A., Hunter, A. L., Donaldson, K., et al. (2013). Global association of air pollution and heart failure: A systematic review and meta-analysis. *Lancet*, 382(9897), 1039–1048.
- Sigma-Aldrich SDS, version 6.1. https://www.sigmaaldrich.com/catalog/product/aldrich/317950?lang=en®ion=CA&gclid=EAIaIQobChMIpTs08-P5AIVGKvsCh2EFwNKEAAAYASAAEgI4_PD_BwE.
- Solvay Solvay specialty polymers, brochure R 10/2017. Version 2.7 B <https://www.solvay.com/sites/g/files/srpend221/files/2018-10/Fomblin-PFPE-Lubes-for-Vacuum-Applications.EN-v2.7.0.pdf>.
- Sorensen, C. M., Gebhart, J., O'Hern, T. J., & Rader, D. J. (2011). Optical measurement techniques: Fundamentals and applications. *Aerosol measurement: Principles, techniques, and applications* (pp. 269–312). (3rd ed.). Hoboken, NJ: Wiley.
- Sousan, S., Koehler, K., Hallett, L., & Peters, T. M. (2016). Evaluation of the Alphasense optical particle counter (OPC-N2) and the Grimm portable aerosol spectrometer (PAS-1.108). *Aerosol Science and Technology*, 50(12), 1352–1365.
- Szymanski, W. W., & Liu, B. Y. H. (1986). On the sizing accuracy of laser optical particle counters. *Particle & Particle Systems Characterization*, 3(1), 1–7.
- Tavakoli, F., & Olfert, J. S. (2013). An instrument for the classification of aerosols by particle relaxation time: Theoretical models of the aerodynamic aerosol classifier. *Aerosol Science and Technology*, 47(8), 916–926.
- Tavakoli, F., Symonds, J. P. R., & Olfert, J. S. (2014). Generation of a monodisperse size-classified aerosol independent of particle charge. *Aerosol Science and Technology*, 48(3), i–iv.
- Tittarelli, A., Borgini, A., Bertoldi, M., Saeger, E. D., Ruprecht, A., Stefanoni, R., et al. (2008). Estimation of particle mass concentration in ambient air using a particle counter. *Atmospheric Environment*, 42(36), 8543–8548.
- Wang, S., & Flagan, R. (1990). Scanning electrical mobility spectrometer. *Journal of Aerosol Science*, 13, 230–240.
- Welker, R. W. (2012). *Developments in surface contamination and cleaning*. Oxford: William Andrew Publishing (Chapter 4) - Size Analysis and Identification of Particles.